

Course Project1_ SG

Sunali Goonesekera

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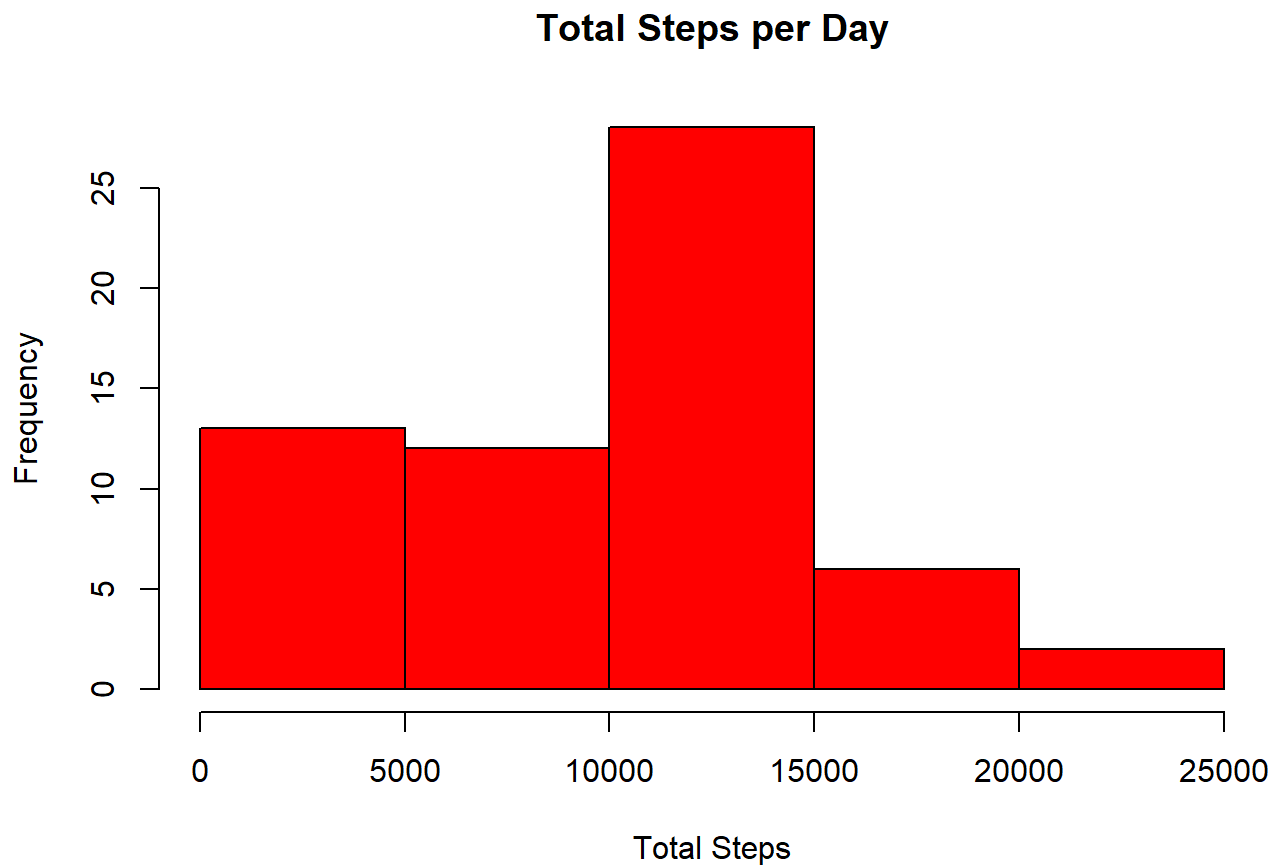
Code used to load the data:

```
library(dplyr)

##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##     filter, lag
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
library(lubridate)
##
## Attaching package: 'lubridate'
## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
activity <- read.csv("C:/Users/sunal/OneDrive/Documents/datasciencecoursera/NewSG/rep_
data_activity/activity.csv", stringsAsFactors=FALSE)
```

#Question 1: Calculating the total number of steps taken per day and plotting a histogram:

```
Steps_by_date <- activity %>%
  group_by(date) %>%
  summarise(total_steps_day = sum(steps, na.rm = TRUE))
hist(Steps_by_date$total_steps_day, main = "Total Steps per Day", xlab = "Total Steps"
, ylab = "Frequency", col = "red")
```



Mean and median total steps per day:

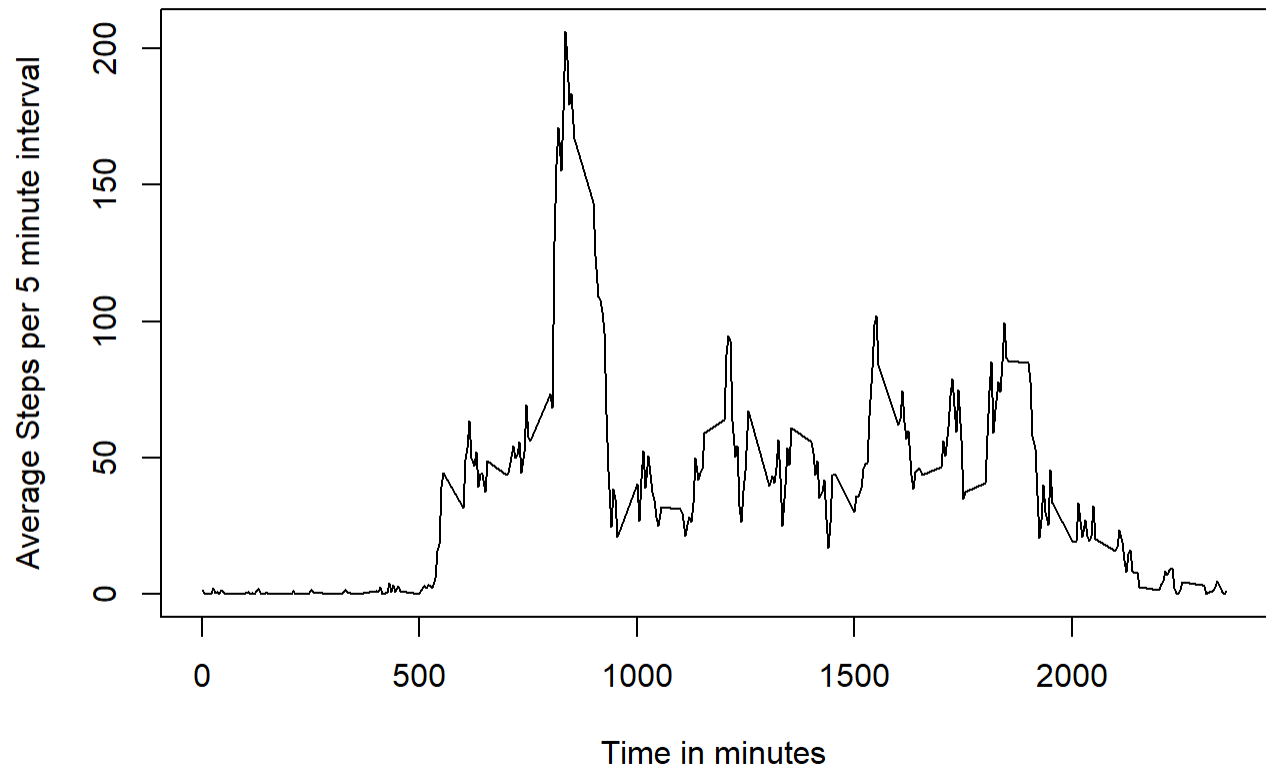
```
names(Steps_by_date)
## [1] "date"          "total_steps_day"
mean_steps <- mean(Steps_by_date$total_steps_day, na.rm = TRUE)
median_steps <- median(Steps_by_date$total_steps_day, na.rm = TRUE)
print(mean_steps)
## [1] 9354.23
print(median_steps)
## [1] 10395
```

#Question 2: Daily average activity pattern and five minute interval with maximum steps:

```
Mean_steps_by_interval <- activity %>%
  group_by(interval) %>%
  summarise(mean_steps = mean(steps, na.rm = TRUE))
plot(x=Mean_steps_by_interval$interval, y=Mean_steps_by_interval$mean_steps, type = "l",
     main = "Average Daily Activity Pattern", xlab = "Time in minutes",
```

```
ylab = "Average Steps per 5 minute interval")
```

Average Daily Activity Pattern

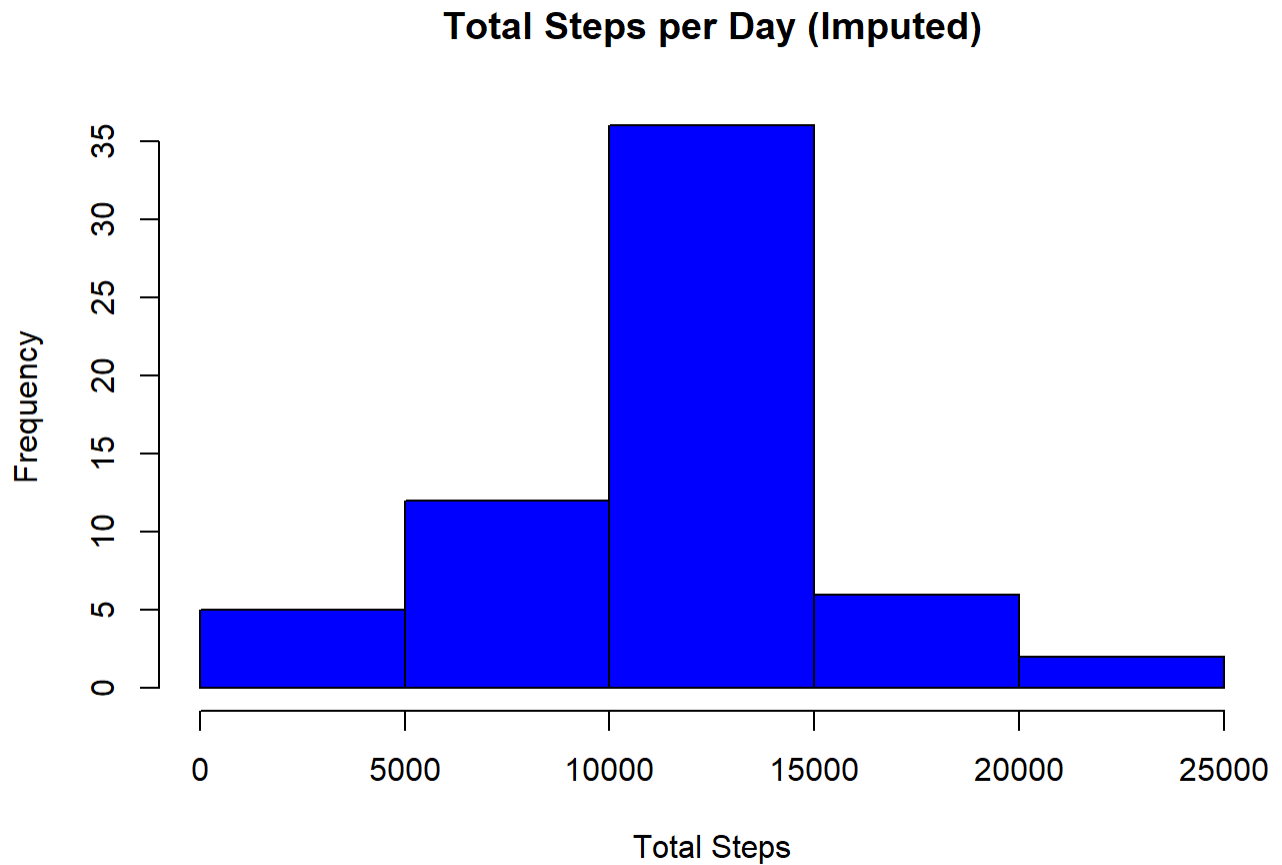


```
max_interval <- Mean_steps_by_interval$interval[which.max(Mean_steps_by_interval$mean_steps)]  
  
print(max_interval)  
## [1] 835
```

#Question 3: Imputing missing values

```
activity_imputed <- activity %>%  
  group_by(interval) %>%  
  mutate(steps = ifelse(is.na(steps), mean(steps, na.rm = TRUE), steps))  
Total_steps_by_date_imputed <- activity_imputed %>%  
  group_by(date) %>%  
  summarise(total_steps = sum(steps))
```

```
hist(Total_steps_by_date_imputed$total_steps, main = "Total Steps per Day (Imputed)",  
xlab = "Total Steps", ylab = "Frequency", col = "blue")
```



```
mean_steps_imputed <- mean(Total_steps_by_date_imputed$total_steps, na.rm = TRUE)  
median_steps_imputed <- median(Total_steps_by_date_imputed$total_steps, na.rm = TRUE)  
print(mean_steps_imputed)  
## [1] 10766.19  
print(median_steps_imputed)  
## [1] 10766.19
```

#Question 4: Are there differences in activity patterns between weekdays and weekends?

```
activity_imputed$weekday <- weekdays(as.Date(activity_imputed$date))  
Weekday <- subset(activity_imputed, weekday %in% c("Monday", "Tuesday", "Wednesday", "  
Thursday", "Friday"))  
Weekend <- subset(activity_imputed, weekday %in% c("Saturday", "Sunday"))  
Mean_steps_by_interval_weekday <- Weekday %>%  
  group_by(interval) %>%
```

```

summarise(mean_steps = mean(steps, na.rm = TRUE))
Mean_steps_by_interval_weekend <- Weekend %>%
  group_by(interval) %>%
  summarise(mean_steps = mean(steps, na.rm = TRUE))
par(mfrow=c(2,1))

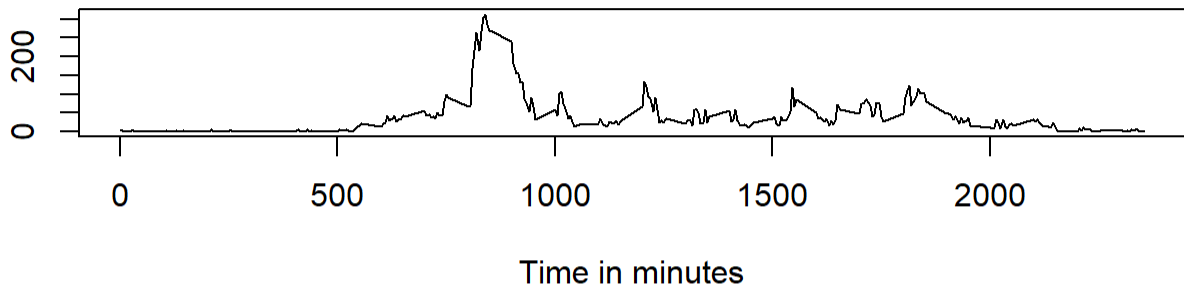
plot(x=Mean_steps_by_interval_weekday$interval, y=Mean_steps_by_interval_weekday$mean_steps, type = "l", main = "Average Daily Activity Pattern (Weekdays)", xlab = "Time in minutes", ylab = "Average Steps per 5 minute interval")

plot(x=Mean_steps_by_interval_weekend$interval, y=Mean_steps_by_interval_weekend$mean_steps, type = "l", main = "Average Daily Activity Pattern (Weekends)", xlab = "Time in minutes", ylab = "Average Steps per 5 minute interval")

```

Average Steps per 5 minute interval

Average Daily Activity Pattern (Weekdays)



Average Daily Activity Pattern (Weekends)

